SpamShield: Reproducible ML Pipeline (Milestone 1)

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GitHub repo: https://github.com/faizan-tnvr004/spamshield-mlops
DagsHub repo: https://dagshub.com/faizan-tnvr004/spamshield-mlops
MLflow tracking server: https://dagshub.com/faizan-tnvr004/spamshield-mlops.mlflow

Overview

SpamShield classifies SMS messages as spam or not spam. This milestone delivers the core ML pipeline only (no Docker, cloud deployment, API or monitoring): data ingestion, validation, preprocessing, training and evaluation of three models, experiment tracking and model registration, all re-runnable with `dvc repro`.

1. MLflow experiment runs

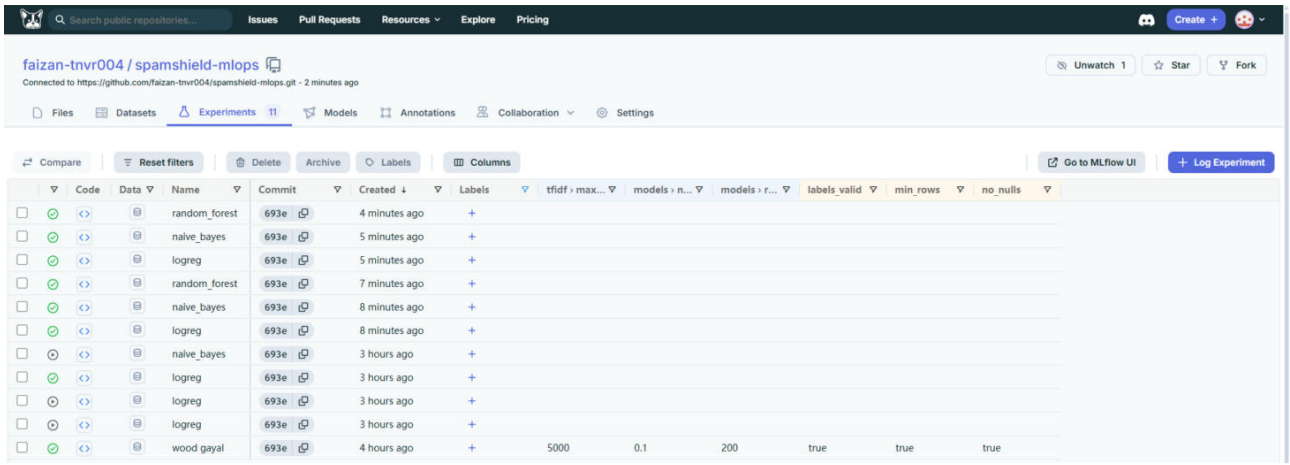


Figure 1: MLflow experiments on DagsHub. Each pipeline run logs three runs (logreg, naive_bayes, random_forest) with parameters, metrics and the model artifact.

Test-set results (20% stratified hold-out):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (baseline)	0.9845	0.9915	0.8855	0.9355	0.9952
Naive Bayes (main)	0.9903	0.9840	0.9389	0.9609	0.9934
Random Forest	0.9884	1.0000	0.9084	0.9520	0.9895

2. Registered model in Staging

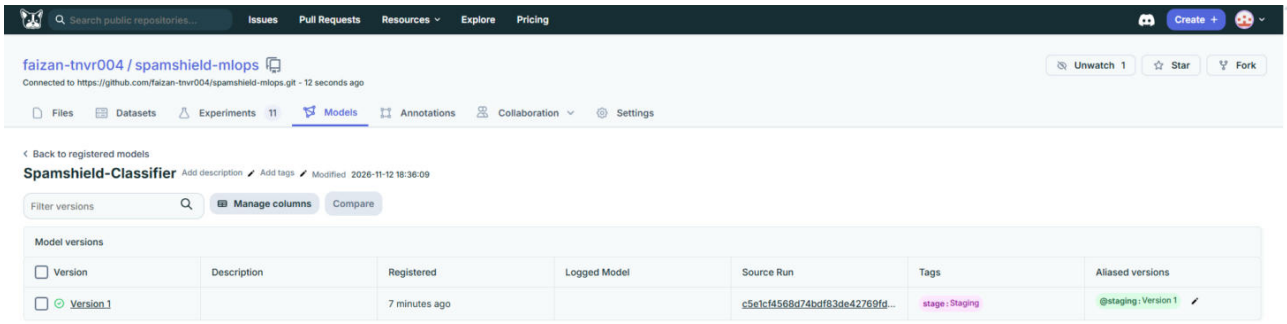
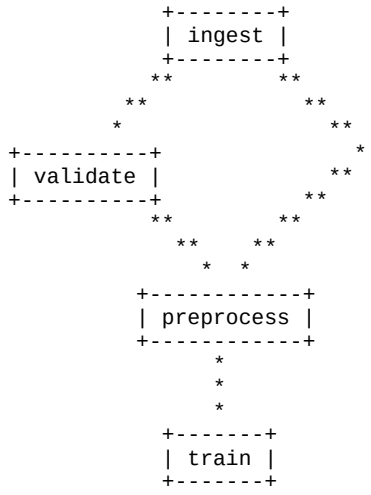


Figure 2: MLflow Model Registry on DagsHub. `spamshield-classifier` *Version 1 (Naive Bayes, best F1) carries the tag stage: Staging and the alias @staging.*

3. DVC pipeline diagram (`dvc dag`)



Write-up

GitHub repository. Organised repo with README (description, setup, run steps, folder structure), `params.yaml`, `dvc.yaml`, `src/` scripts. `.gitignore` excludes raw/processed data, model weights, local MLflow stores and credentials.

Data pipeline and versioning. `ingest.py` downloads the UCI SMS Spam Collection (5,572 messages) from its source. `validate.py` checks schema, nulls, label set, text-length range, minimum row count and spam ratio, and fails the pipeline if any check fails. `preprocess.py` lowercases, normalises URLs and numbers, de-duplicates and makes a stratified train/test split. Data and models are versioned with DVC and pushed to the DagsHub DVC remote.

Model core. TF-IDF (unigrams and bigrams, 5,000 features) feeds three scikit-learn classifiers: Logistic Regression as the baseline, then Naive Bayes and Random Forest. All are evaluated on the same held-out set with accuracy, precision, recall, F1 and ROC-AUC. Naive Bayes gives the best F1 (0.961) and recall (0.939).

Experiment tracking. `train.py` logs hyperparameters, metrics, `params.yaml` and the fitted pipeline for every run to the remote MLflow server on DagsHub. The best-F1 model is registered as `spamshield-classifier` and set to Staging. Credentials are read from environment variables, never committed.

DVC pipeline. `dvc.yaml` defines four named stages (ingest, validate, preprocess, train) with explicit dependencies, parameters, outputs and metrics. `dvc repro` rebuilds everything from scratch, and changing a parameter re-runs only the affected stages.